QF620 Stochastic Modelling in Finance — Session 8 Reference Cheatsheet

Static Replication of European Payoffs: stochastic-vol parameters and the smile, Leibniz's rule, model-free pricing, Breeden–Litzenberger, Carr–Madan, log contracts, variance swaps and VIX, digital options

Built from Session 8 slides (Tee Chyng Wen, SMU) · October 2026 · revised edition with intuition and full slide derivations

Contents

Notation used throughout (slide conventions) **2**

0. Topic map **2**

1 Concept entries **3**

1.1 Correlation ρ: distribution and pricing [S2–S4] . . . 3
1.2 Vol-of-vol ν: distribution and pricing [S5–S7] . . . 3
1.3 Differentiating integrals; Leibniz's rule [S8–S9] . . . 3
1.4 The model-free framework [S10–S11] . . . 4
1.5 Implied risk-neutral density: motivation [S12–S13] . . . 4
1.6 Breeden–Litzenberger (1978) [S14–S15] . . . 4
1.7 Static replication: setup [S16] . . . 5
1.8 Integration by parts twice [S17–S18] . . . 5
1.9 Carr–Madan static replication formula [S18–S19] . . . 6
1.10 Example: power payoff aS_T^b [S20] . . . 7
1.11 Log contract [S21] . . . 7
1.12 Variance swaps: contract [S22] . . . 7
1.13 Variance swaps: pricing [S23–S25] . . . 8
1.14 VIX index [S26] . . . 8
1.15 Payoffs with discontinuities [S27–S28] . . . 9
1.16 Digital via call spread [S29] . . . 9
1.17 Implementation *[Beyond slides]* (tested) . . . 9

2 Comparison tables **10**

2.1 What ρ and ν do [S2–S7] . . . 10
2.2 Replication weights by payoff . . . 10
2.3 Static vs dynamic replication . . . 10

3 Master formula sheet **10**

4 Glossary **10**

5 Self-test **11**

Questions . . . 11
Solutions . . . 11

Appendix A. Slide coverage checklist **12**

Appendix B. Flag summary **12**

Source: "Session 8: Static Replication of European Payoffs" (Tee Chyng Wen, SMU, 29 slides).

How to read this sheet. Each concept entry has up to five parts: **The idea in words** (what the concept is and why it is there), the formula with a shaded where box naming every symbol, **Derivation** (the slide's argument line by line, with the reason for each step in brown), **Worked example**, and **Exam angle** (traps and what to be able to reproduce). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged *[Beyond slides]*. △! marks a slide ambiguity or error, with the corrected version.

What you must be able to do after this session: explain what ρ and ν do to the distribution and the smile; state and use Leibniz's rule; derive Breeden–Litzenberger; derive the Carr–Madan formula by integrating by parts twice and using put–call parity; replicate power and log payoffs; derive the variance-swap fair strike; value a digital by static replication and approximate it with a call spread.

Notation used throughout (slide conventions)

Symbol	Meaning	Notes
$\rho,\ \nu$	spot–vol correlation; vol-of-vol (SABR/Heston)	[S2–S7]
$f(s)$	risk-neutral density of S_T	[S10, S14]
$C(K),\ P(K)$	European call / put price with strike K, expiry T	[S10, S15]
F	forward S_0e^{rT} (on [S8–S9] F is an antiderivative)	[S16]
$h(S_T)$	European payoff; h', h'' its derivatives	[S16]
V_0	today's value of the payoff	[S16]
$\sigma_R^2,\ \sigma_K^2$	realized variance; strike variance	[S22]
σ_t	(possibly stochastic) instantaneous volatility	[S23]
$\bar{K}$	digital strike	[S28]

0. Topic map

```
PREREQUISITES: S6 smile, SABR/Heston ; S4 Itô ; S5/S7 risk-neutral pricing, BS call ; calculus (by parts)
    ▼
SESSION 8├─
 STOCH-VOL PARAMETERS [–S2S7]|
     ├─ ρ < 0: fat left tail ⇒ OTM puts up, calls down (skew)                  [–S2S4]|
     └─ ν ↑: kurtosis, both tails fat ⇒ OTM puts and calls up (curvature)       [–S5S7]├─
 LEIBNIZ [–S8S9]|
     └─ d/dx ∫_{u(x)}^{v(x)} f(x,t)dt = f(x,v)′v − f(x,u)′u + ∫∂f∂/x dt          [–S8S9]├─
 MODEL-FREE & –BREEDENLITZENBERGER [–S10S15]|
     ├─ C(K) = e–^{rT∫}_K∞^ (–sK) f(s) ds ; extract f from many quotes           [–S10S13]|
     └─ ∂C∂/K = –e–^{rT∫}_K∞^ f ; ∂²C∂/K² = ∂²P∂/K² = e–^{rT} f(K)               [–S14S15]├─
 –CARRMADAN [–S16S21]|
     ├─ ₀V = ∫_0^F h ″P dK + ∫_F∞^ h ″C dK ; by parts twice ; parity            [–S16S18]|
     ├─ ₀V = e–^{rT}h(F) + ∫_0^F ″hP dK + ∫_F∞^ ″hC dK                          [S19]|
     └─ examples: a S_T^b ; log contract                                       [–S20S21]├─
 VARIANCE SWAPS & VIX [–S22S26]|
     ├─ σ²dt = 2dS/S – 2d log S ⇒ ∫σE²dt = 2rT – 2E log(S_T/₀S)                [–S22S24]|
     └─ E∫σ*²dt = 2e^{rT∫}[_0^F P/K² + ∫_F∞^ C/K²] ; VIX                       [–S25S26]└─
 DISCONTINUOUS PAYOFFS [–S27S29]
      └─ digital = –∂C(K)∂/K ≈ call spread                                      [–S27S29]
```

The storyline of the session.

- Stochastic-vol parameters reshape the risk-neutral distribution (ρ skews it, ν fattens both tails). The smile is just the market's picture of that distribution.
- If calls trade at every strike, differentiating $C(K)$ twice in K recovers the density with no model (**Breeden–Litzenberger**). Leibniz's rule is the tool.
- Any European payoff's price is an integral of the density. Substitute $f = e^{rT}\partial^2 C/\partial K^2$ and integrate by parts twice: the payoff becomes a bond + forwards + a strip of OTM options (**Carr–Madan**). This hedge is *static*: set up once, never rebalanced, no model.

- Itô turns realized variance into $2\frac{\mathrm{d}S}{S} - 2\,\mathrm{d}\log S$: a dynamic position worth $2rT$ plus a log contract, which is replicated statically with $1/K^2$-weighted options. That is how variance swaps and the VIX are priced.

1 Concept entries

1.1 Correlation ρ: distribution and pricing [S2–S4]

The idea in words
With $\rho < 0$, when the stock falls volatility rises, so down-moves come in bigger, wilder steps and the left tail spreads out; up-moves happen in a calm regime and the right tail stays thin. Result: negative skew. OTM puts pay off in the left tail, so they become more valuable; OTM calls lose. In implied-vol terms the smile slopes down. $\rho > 0$ is the mirror image.

- ρ controls the **skewness** of stock returns [S2].
- $\rho < 0$: fat left tail, thin right tail $\Rightarrow$ OTM puts up, OTM calls down relative to Black–Scholes [S2–S3].
- $\rho > 0$: fat right tail, thin left tail $\Rightarrow$ positive skew [S2–S3].
- Smile picture [S4]: $\rho = -0.5$ implied vol falls with strike; $\rho = 0$ roughly symmetric; $\rho = +0.5$ tilts up.

Exam angle
△! [S2] says ρ is "proportional" to skewness; the relation is monotone (higher ρ, more positive skew), not literally proportional.

1.2 Vol-of-vol ν: distribution and pricing [S5–S7]

The idea in words
Random volatility mixes calm-regime normals (narrow) with turbulent-regime normals (wide). A mixture of normals with different widths has a sharper peak and fatter tails than one normal: higher kurtosis. Both tails gain, so both OTM puts and OTM calls get dearer: the smile acquires curvature (both wings lift). Without correlation, ν adds kurtosis only; skew comes from ρ.

- $\nu = 0$: volatility is deterministic [S6].
- $\nu > 0$ increases kurtosis: two fat tails [S5–S6] $\Rightarrow$ raises both OTM put and call prices [S5, S7].
- If $\rho = 0$, raising ν only adds kurtosis: far-from-the-money options gain relative to near-the-money [S5].
- Smile picture [S7]: larger ν lifts both wings.

Exam angle
△! [S5] "back to a normal log-return distribution (if $\beta = 0$)" should read $\beta = 1$; with $\beta = 0$, $\nu = 0$ the returns are normal in price units (Bachelier).

1.3 Differentiating integrals; Leibniz's rule [S8–S9]

The idea in words
An integral $\int_{u(x)}^{v(x)} f(x,t)\,\mathrm{d}t$ depends on x in three places: the top limit, the bottom limit, and the integrand. Its derivative collects one term for each. Moving the top limit up by $\mathrm{d}v$ adds a thin sliver of area of height f at the top; moving the bottom limit up removes one at the bottom; and changing the integrand changes the area everywhere in between. You need this to differentiate an option price with respect to its strike, which appears both as a limit and inside the payoff.

Derivation [S8–S9]
Chain rule for $g(u(x), v(x))$: $\frac{\mathrm{d}g}{\mathrm{d}x} = \frac{\partial g}{\partial u}u'(x) + \frac{\partial g}{\partial v}v'(x)$ [S8]. Let F be an antiderivative of f in t:

$$I(v,u,x) = \int_{u(x)}^{v(x)} f(x,t)\,\mathrm{d}t = F(x,v(x)) - F(x,u(x)) \qquad \text{(fundamental theorem)}$$

$$\frac{\partial I}{\partial v} = f(x,v(x)), \qquad \frac{\partial I}{\partial u} = -f(x,u(x)) \qquad (\partial_t F = f)$$

$$\frac{\mathrm{d}I}{\mathrm{d}x} = \frac{\partial I}{\partial v}v'(x) + \frac{\partial I}{\partial u}u'(x) + \frac{\partial I}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}x} \qquad \text{(chain rule, 3 arguments)}$$

$$= f(x,v(x))v'(x) - f(x,u(x))u'(x) + \int_{u(x)}^{v(x)} \frac{\partial f(x,t)}{\partial x}\,\mathrm{d}t \qquad \text{([S9])}$$

Special case $f = f(t)$: $\frac{\mathrm{d}}{\mathrm{d}x}\int_{u(x)}^{v(x)} f(t)\,\mathrm{d}t = f(v(x))v'(x) - f(u(x))u'(x)$.

where

$f(x,v)v'$	moving the upper limit adds a sliver of height f at the top.
$-f(x,u)u'$	moving the lower limit removes a sliver at the bottom.
$\int \partial_x f\,\mathrm{d}t$	the integrand itself changing with x (zero if $f = f(t)$ only).

1.4 The model-free framework [S10–S11]

The idea in words
All European prices are discounted risk-neutral expectations, i.e. integrals against the risk-neutral density f. Usually we pick a model (Black–Scholes, Bachelier, DD, SABR), calibrate it, and the model dictates f. But if options trade at many strikes, the market prices themselves pin down f. Skip the model and read f off the prices directly.

$$C(K) = e^{-rT}\mathbb{E}^*[(S_T - K)^+] = e^{-rT}\int_K^\infty (s-K)f(s)\,\mathrm{d}s.$$

where

$f(s)$	risk-neutral density of S_T, not specified by a model.
$\int_K^\infty$	the payoff is zero for $s < K$.

Model-based [S10]: e.g. Black–Scholes $C(K) = \frac{e^{-rT}}{\sqrt{2\pi}}\int_{x^*}^\infty (S_0 e^{(r-\frac{\sigma^2}{2})T+\sigma\sqrt{T}x} - K)e^{-x^2/2}\,\mathrm{d}x$; once calibrated, the density is fixed. Model-free [S11]: with enough liquid quotes $C(K_1), C(K_2), \ldots$ extract f directly.

1.5 Implied risk-neutral density: motivation [S12–S13]

- Black–Scholes assumes $\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t^*$ under $\mathbb{Q}^*$ with one parameter σ [S12].
- Vanilla options are liquid; their prices are observed. Inverting gives one implied vol per strike and maturity [S12–S13].
- Next level: for a given maturity, find the implied *distribution* that matches the whole smile. The tool is Leibniz's rule [S13].

1.6 Breeden–Litzenberger (1978) [S14–S15]

The idea in words
Differentiate the call price with respect to strike. The first derivative is (minus, discounted) the probability of finishing above K: raising the strike by 1 costs you \$1 in every in-the-money scenario. The second derivative is the density at K. Intuition: a butterfly $C(K-\Delta) - 2C(K) + C(K+\Delta)$ pays a small tent around K, so its price is roughly "probability of landing near K" $\times$ discount. Prices of options across strikes are therefore a complete map of the risk-neutral distribution.

First and second strike derivatives of the call [S14]
Apply Leibniz with $x = K$, $u(K) = K$ ($u' = 1$), $v = \infty$ ($v' = 0$), integrand $(s-K)f(s)$:

$$\frac{\partial C}{\partial K} = e^{-rT}\Big[\lim_{x\to\infty}(x-K)f(x)\frac{\mathrm{d}x}{\mathrm{d}K} - (K-K)f(K)\frac{\mathrm{d}K}{\mathrm{d}K} + \int_K^\infty \frac{\partial(s-K)}{\partial K}f(s)\,\mathrm{d}s\Big] \quad \text{(Leibniz)}$$

$$= e^{-rT}\Big[0 - 0 - \int_K^\infty f(s)\,\mathrm{d}s\Big] = -e^{-rT}\int_K^\infty f(s)\,\mathrm{d}s \quad (\tfrac{\mathrm{d}x}{\mathrm{d}K} = 0;\ \text{payoff 0 at } K)$$

$$\frac{\partial^2 C}{\partial K^2} = -e^{-rT}\Big[\lim_{x\to\infty} f(x)\frac{\mathrm{d}x}{\mathrm{d}K} - f(K)\frac{\mathrm{d}K}{\mathrm{d}K} + \int_K^\infty \frac{\partial f(s)}{\partial K}\,\mathrm{d}s\Big] \quad \text{(Leibniz again)}$$

$$= -e^{-rT}\big[0 - f(K) + 0\big] = e^{-rT}f(K) \quad (f(s)\ \text{has no } K)$$

Same for puts [S15] *[Beyond slides]* (the slide states the result)
$P(K) = e^{-rT}\int_0^K (K-s)f(s)\,\mathrm{d}s$. Leibniz with upper limit K: $\frac{\partial P}{\partial K} = e^{-rT}\big[(K-K)f(K)\cdot 1 + \int_0^K f(s)\,\mathrm{d}s\big] = e^{-rT}\int_0^K f(s)\,\mathrm{d}s$, and $\frac{\partial^2 P}{\partial K^2} = e^{-rT}f(K)$.

$$\boxed{\frac{\partial^2 C(K)}{\partial K^2} = \frac{\partial^2 P(K)}{\partial K^2} = e^{-rT}f(K)} \qquad \text{(Breeden–Litzenberger, 1978)}$$

where

$-e^{rT}\frac{\partial C}{\partial K}$	$= \mathbb{Q}^*(S_T > K)$ *[Beyond slides]* (price of a digital, see 1.15).
$f(K)$	the implied distribution: curvature of price in strike.

- *[Beyond slides]* Butterfly: $\frac{C(K-\Delta)-2C(K)+C(K+\Delta)}{\Delta^2} \approx e^{-rT}f(K)$; convexity of C in K is a no-arbitrage condition ($f \geq 0$).
- **Carr–Madan (1998):** any European payoff can be replicated using cash, forwards and European calls and puts [S15].

1.7 Static replication: setup [S16]

The idea in words
The value of any payoff $h(S_T)$ is $e^{-rT}\int hf$. Replace $e^{-rT}f(K)$ by the second strike-derivative of option prices: below the forward use puts, above use calls (these are the liquid OTM options). The rest is calculus: integrate by parts twice to move the derivatives off the option prices and onto h.

For twice-differentiable h, with $F = S_0 e^{rT}$:

$$V_0 = e^{-rT}\mathbb{E}[h(S_T)] = e^{-rT}\int_0^\infty h(s)f(s)\,\mathrm{d}s = \underbrace{\int_0^F h(K)\frac{\partial^2 P(K)}{\partial K^2}\,\mathrm{d}K}_{(1)} + \underbrace{\int_F^\infty h(K)\frac{\partial^2 C(K)}{\partial K^2}\,\mathrm{d}K}_{(2)}.$$

where

$e^{-rT}f(K)$	replaced by P'' below F and C'' above (Breeden–Litzenberger); discount absorbed.
$s \to K$	dummy variable renamed as a reminder the derivatives are in strike.
split at F	uses liquid OTM/ATM options: low-strike puts, high-strike calls.

1.8 Integration by parts twice [S17–S18]

Call integral (2) [S17] — every line
By parts: $\int hC'' = [hC'] - \int h'C'$, then $\int h'C' = [h'C] - \int h''C$.

$$\begin{aligned}
\int_F^\infty h(K)\frac{\partial^2 C}{\partial K^2}\,\mathrm{d}K &= \Big[h(K)\frac{\partial C}{\partial K}\Big]_F^\infty - \int_F^\infty h'(K)\frac{\partial C}{\partial K}\,\mathrm{d}K && \text{(by parts, 1st time)}\\
&= \Big[h(K)\frac{\partial C}{\partial K}\Big]_F^\infty - \Big[h'(K)C(K)\Big]_F^\infty + \int_F^\infty h''(K)C(K)\,\mathrm{d}K && \text{(by parts, 2nd time)}\\
&= \Big[0 - h(F)\frac{\partial C(F)}{\partial K}\Big] - \Big[0 - h'(F)C(F)\Big] + \int_F^\infty h''C\,\mathrm{d}K && (C(\infty)=C'(\infty)=0)\\
&= -h(F)\frac{\partial C(F)}{\partial K} + h'(F)C(F) + \int_F^\infty h''(K)C(K)\,\mathrm{d}K
\end{aligned}$$

Put integral (1) [S18] ***[Beyond slides]*** **(same steps; $P(0)=P'(0)=0$)**

$$\int_0^F h\frac{\partial^2 P}{\partial K^2}\,\mathrm{d}K = \Big[h\frac{\partial P}{\partial K}\Big]_0^F - \Big[h'P\Big]_0^F + \int_0^F h''P\,\mathrm{d}K = h(F)\frac{\partial P(F)}{\partial K} - h'(F)P(F) + \int_0^F h''(K)P(K)\,\mathrm{d}K$$

Combining (1) + (2) [S18]

$$V_0 = h(F)\Big[-\frac{\partial C(F)}{\partial K} + \frac{\partial P(F)}{\partial K}\Big] + h'(F)\big[C(F)-P(F)\big] + \int_0^F h''(K)P(K)\,\mathrm{d}K + \int_F^\infty h''(K)C(K)\,\mathrm{d}K.$$

1.9 Carr–Madan static replication formula [S18–S19]

The idea in words

Put–call parity cleans up the two boundary terms. Its strike-derivative says $C'-P' = -e^{-rT}$, so the first bracket is e^{-rT}: a zero-coupon bond paying $h(F)$. At $K=F$ the call and put are worth the same, so the second bracket vanishes (it corresponds to $h'(F)$ forwards, which are worth zero today but are part of the hedge). What remains: bond + a strip of OTM puts and calls, each weighted by the payoff's curvature $h''(K)$. Convex payoffs mean long options, concave payoffs mean short options, linear payoffs need no options at all.

Using put–call parity [S18–S19]

$$\begin{aligned}
C(K)-P(K) &= S_0 - Ke^{-rT} && \text{(put–call parity)}\\
\frac{\partial C}{\partial K} - \frac{\partial P}{\partial K} &= -e^{-rT} \;\Rightarrow\; -\frac{\partial C(F)}{\partial K} + \frac{\partial P(F)}{\partial K} = e^{-rT} && \text{(differentiate in } K)\\
C(F)-P(F) &= S_0 - Fe^{-rT} = S_0 - S_0 e^{rT}e^{-rT} = 0 && (K=F)
\end{aligned}$$

$$\boxed{V_0 = e^{-rT}h(F) + \int_0^F h''(K)P(K)\,\mathrm{d}K + \int_F^\infty h''(K)C(K)\,\mathrm{d}K}$$

where

$e^{-rT}h(F)$	zero-coupon bond paying $h(F)$.
forward term	$h'(F)[C(F)-P(F)]$: $h'(F)$ forwards struck at F; worth zero today, so absent from the price but held in the hedge *[Beyond slides]*.
$h''(K)P(K)$	hold $h''(K)\,\mathrm{d}K$ OTM puts at each strike $K<F$.
$h''(K)C(K)$	hold $h''(K)\,\mathrm{d}K$ OTM calls at each strike $K>F$.

[Beyond slides] Payoff-level identity: $h(S_T)=h(F)+h'(F)(S_T-F)+\int_0^F h''(K)(K-S_T)^+\,\mathrm{d}K+\int_F^\infty h''(K)(S_T-K)^+\,\mathrm{d}K$ (Taylor with integral remainder). The hedge is static and model-free: only today's option prices are needed.

1.10 Example: power payoff aS_T^b [S20]

[S20]
$h(S)=aS^b$, $h'=abS^{b-1}$, $h''=ab(b-1)S^{b-2}$; $h(F)=aF^b$. Plug into Carr–Madan:

$$V_0=e^{-rT}aF^b+\int_0^F ab(b-1)K^{b-2}P(K)\,\mathrm{d}K+\int_F^\infty ab(b-1)K^{b-2}C(K)\,\mathrm{d}K.$$

where

$ab(b-1)K^{b-2}$	option weight: positive for $b>1$ (convex, long options); negative for $0<b<1$ (concave, short); zero for $b=1$ (just a forward).
check	*[Beyond slides]* under Black–Scholes $V_0=e^{-rT}aS_0^b\exp[b(r-\frac{1}{2}\sigma^2)T+\frac{1}{2}b^2\sigma^2T]$ (MGF).

1.11 Log contract [S21]

(1) Black–Scholes value (worked here)
Under $\mathbb{Q}^*$, $\log\frac{S_T}{S_0}=(r-\frac{1}{2}\sigma^2)T+\sigma W_T^*$, so $\mathbb{E}^*[\log\frac{S_T}{S_0}]=(r-\frac{1}{2}\sigma^2)T$ and $V_0=e^{-rT}(r-\frac{1}{2}\sigma^2)T$.

(2) Carr–Madan replication (worked here)
$h(K)=\log\frac{K}{S_0}$, $h'=\frac{1}{K}$, $h''=-\frac{1}{K^2}$; $h(F)=\log\frac{S_0e^{rT}}{S_0}=rT$:

$$V_0=e^{-rT}rT-\int_0^F\frac{P(K)}{K^2}\,\mathrm{d}K-\int_F^\infty\frac{C(K)}{K^2}\,\mathrm{d}K.$$

Short a strip of OTM options with weight $1/K^2$ (the log is concave). Equating (1) and (2) under Black–Scholes: $\int_0^F\frac{P}{K^2}\,\mathrm{d}K+\int_F^\infty\frac{C}{K^2}\,\mathrm{d}K=e^{-rT}\frac{\sigma^2T}{2}$.

1.12 Variance swaps: contract [S22]

The idea in words
A variance swap pays the difference between realized variance and a fixed strike. It gives pure exposure to volatility, with no need to delta- or gamma-hedge vanilla options. Its fair strike is the risk-neutral expectation of realized variance, and the next entry shows that expectation is a model-free strip of options.

$$\text{Payoff}=\text{Notional}\times(\sigma_R^2-\sigma_K^2),\qquad \sigma_R^2=\frac{252}{N}\sum_{i=1}^{N}\left(\log\frac{S_{t_i}}{S_{t_{i-1}}}\right)^2.$$

where

σ_R^2	realized annualized variance from daily log returns (zero mean assumed).
σ_K^2	strike variance; the fair strike makes the swap worth zero at inception.
252, N	trading days per year; days in the contract.

1.13 Variance swaps: pricing [S23–S25]

The idea in words

Itô's formula is the bridge. For $\log S$, the $\mathrm{d}t$ term is $-\frac{1}{2}\sigma_t^2$, so variance equals "twice the return $\mathrm{d}S/S$ minus twice the log-return". The $\mathrm{d}S/S$ part is a continuously rebalanced position whose risk-neutral value is simply $2rT$; the log part is a log contract held to expiry, which Carr–Madan replicates statically. This works for *any* volatility process σ_t: model-free.

From Itô to options [S23–S25]

$\sum_i \log^2 \frac{S_{t_i}}{S_{t_{i-1}}} \approx \int_0^T \sigma_t^2\,\mathrm{d}t$ [S23]. For $\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma_t S_t\,\mathrm{d}W_t^*$:

$$\mathrm{d}\log S_t = \frac{\mathrm{d}S_t}{S_t} - \frac{1}{2}\frac{(\mathrm{d}S_t)^2}{S_t^2} = \frac{\mathrm{d}S_t}{S_t} - \frac{1}{2}\sigma_t^2\,\mathrm{d}t \qquad \text{(Itô, } f'' = -1/S^2\text{)}$$

$$\sigma_t^2\,\mathrm{d}t = 2\Big(\frac{\mathrm{d}S_t}{S_t} - \mathrm{d}\log S_t\Big) \qquad \text{(rearrange)}$$

$$\int_0^T \sigma_t^2\,\mathrm{d}t = 2\int_0^T \frac{\mathrm{d}S_t}{S_t} - 2\log\frac{S_T}{S_0} \qquad \text{(integrate)}$$

$$\mathbb{E}^*\Big[\int_0^T \sigma_t^2\,\mathrm{d}t\Big] = 2\mathbb{E}^*\Big[\int_0^T \frac{\mathrm{d}S_t}{S_t}\Big] - 2\mathbb{E}^*\Big[\log\frac{S_T}{S_0}\Big] \qquad \text{(take } \mathbb{E}^*\text{)}$$

$$2\mathbb{E}^*\Big[\int_0^T \frac{rS_t\,\mathrm{d}t + \sigma_t S_t\,\mathrm{d}W_t^*}{S_t}\Big] = 2rT \qquad \Big(\mathbb{E}^*\textstyle\int \sigma\,\mathrm{d}W^* = 0\Big)$$

$$2\mathbb{E}^*\Big[\log\frac{S_T}{S_0}\Big] = 2\log\frac{F}{S_0} - 2e^{rT}\int_0^F \frac{P}{K^2}\,\mathrm{d}K - 2e^{rT}\int_F^\infty \frac{C}{K^2}\,\mathrm{d}K \qquad (1.11\ \times 2e^{rT})$$

$$= 2rT - 2e^{rT}\int_0^F \frac{P}{K^2}\,\mathrm{d}K - 2e^{rT}\int_F^\infty \frac{C}{K^2}\,\mathrm{d}K \qquad \Big(\log\frac{F}{S_0} = rT\Big)$$

Subtract: the $2rT$ terms cancel.

$$\boxed{\mathbb{E}^*\Big[\int_0^T \sigma_t^2\,\mathrm{d}t\Big] = 2e^{rT}\Big[\int_0^F \frac{P(K)}{K^2}\,\mathrm{d}K + \int_F^\infty \frac{C(K)}{K^2}\,\mathrm{d}K\Big], \qquad \sigma_K^2 = \frac{1}{T}\mathbb{E}^*\Big[\int_0^T \sigma_t^2\,\mathrm{d}t\Big]\ \textit{[Beyond slides]}}$$

where

$2\int \mathrm{d}S/S$	continuously rebalanced position holding $2/S_t$ shares (dynamic), worth $2rT$.
$-2\log(S_T/S_0)$	short two log contracts (static).
$2e^{rT}$	forward value: scales today's option prices to time T.
$1/K^2$	weights: much more on low-strike puts than high-strike calls ⇒ skew-sensitive [S25].

Exam angle

△! [S25] writes $\mathbb{E}$ without the $*$; it is the risk-neutral expectation. Sanity check *[Beyond slides]*: under Black–Scholes, using 1.11, the strip equals $e^{-rT}\frac{\sigma^2 T}{2}$, so $\mathbb{E}^*\int\sigma^2\,\mathrm{d}t = \sigma^2 T$ and $\sigma_K = \sigma$ ✓.

1.14 VIX index [S26]

$$\sigma^2 = \frac{2}{T}\sum_i \frac{\Delta K_i}{K_i^2}e^{RT}Q(K_i) - \frac{1}{T}\Big[\frac{F}{K_0} - 1\Big]^2, \qquad \mathrm{VIX} = 100\,\sigma.$$

where

$Q(K_i)$	mid of the OTM option at K_i: call if $K_i > K_0$, put if $K_i < K_0$, average at K_0.
K_0	first strike below the forward F (the discrete split point).
ΔK_i	$(K_{i+1} - K_{i-1})/2$; one-sided at the ends.
first term	discrete Riemann sum of $\frac{2e^{rT}}{T}[\int_0^F \frac{P}{K^2} + \int_F^\infty \frac{C}{K^2}]$ from 1.13.
correction	*[Beyond slides]* adjusts for splitting at $K_0 \neq F$; zero if $K_0 = F$.

1.15 Payoffs with discontinuities [S27–S28]

The idea in words
Carr–Madan needs h'', which does not exist at a jump. So go back one step: start from $\int h\,C''$ (or P'') and integrate by parts on the pieces where h is smooth. For a digital call the payoff is flat (1) beyond the strike, so after integration by parts only one boundary term survives: minus the slope of the call price at $\bar{K}$.

Start from $V_0 = e^{-rT}\int_0^\infty h(K)f(K)\,\mathrm{d}K = \int_0^F h\frac{\partial^2 P}{\partial K^2}\,\mathrm{d}K + \int_F^\infty h\frac{\partial^2 C}{\partial K^2}\,\mathrm{d}K$ and integrate by parts twice on each smooth piece [S27].

Cash-or-nothing digital call [S28], $h = \mathbb{1}_{\{S_T \geq \bar{K}\}}$, $\bar{K} \geq F$
$h = 0$ on $[0, \bar{K})$, so only $\int_{\bar{K}}^\infty$ survives; there $h = 1$, $h' = h'' = 0$:

$$V_0 = \int_{\bar{K}}^\infty h(K)\frac{\partial^2 C}{\partial K^2}\,\mathrm{d}K = \Big[h\frac{\partial C}{\partial K}\Big]_{\bar{K}}^\infty - \Big[h'C\Big]_{\bar{K}}^\infty + \int_{\bar{K}}^\infty h''C\,\mathrm{d}K \qquad \text{(by parts twice)}$$

$$= \Big[0 - 1\cdot\frac{\partial C(\bar{K})}{\partial K}\Big] - 0 + 0 = -\frac{\partial C(\bar{K})}{\partial K} \qquad (C'(\infty) = 0)$$

By Breeden–Litzenberger this is $e^{-rT}\int_{\bar{K}}^\infty f = e^{-rT}\mathbb{Q}^*(S_T > \bar{K})$; under Black–Scholes $e^{-rT}\Phi(d_2)$ *[Beyond slides]*.

1.16 Digital via call spread [S29]

$$-\frac{\partial C(\bar{K})}{\partial K} \approx \frac{C(\bar{K} - \Delta K) - C(\bar{K})}{\Delta K}.$$

where

$1/\Delta K$	long $1/\Delta K$ calls at $\bar{K} - \Delta K$, short $1/\Delta K$ calls at $\bar{K}$.
payoff	0 below $\bar{K} - \Delta K$, linear ramp to 1 at $\bar{K}$, then 1: converges to the digital as $\Delta K \to 0$ (slide: $\bar{K} = 50$, $\Delta K = 0.10, 0.05, 0.01$).

[Beyond slides] This one-sided spread pays at least the digital (super-replicates); in practice the spread is limited by strike spacing and the large notional $1/\Delta K$.

1.17 Implementation *[Beyond slides]* (tested)

```
import numpy as np
from scipy.stats import norm
from scipy.integrate import quad
S0, r, sigma, T = 100.0, 0.05, 0.2, 1.0
F, D = S0*np.exp(r*T), np.exp(-r*T)
def call(K):
    d1 = (np.log(S0/K) + (r + 0.5*sigma**2)*T) / (sigma*np.sqrt(T))
    return S0*norm.cdf(d1) - K*D*norm.cdf(d1 - sigma*np.sqrt(T))
def put(K):                                  # put-call parity
    return call(K) - S0 + K*D
# 1) Breeden-Litzenberger: f(K) = e^{rT} d2C/dK2 (central difference)
K, h = 110.0, 0.01
f_BL = (call(K + h) - 2*call(K) + call(K - h)) / h**2 / D
m, s = np.log(S0) + (r - 0.5*sigma**2)*T, sigma*np.sqrt(T)
print(round(f_BL, 6), round(norm.pdf((np.log(K) - m)/s)/(K*s), 6))   # 0.017192 0.017192
# 2) Carr-Madan: V0 = D h(F) + int_0^F h''P dK + int_F^inf h''C dK
def carr_madan(h, h2):
    return (D*h(F) + quad(lambda k: h2(k)*put(k), 1e-8, F, limit=200)[0]
                   + quad(lambda k: h2(k)*call(k), F, np.inf, limit=200)[0])
a, b = 1.0, 2.0                              # payoff a S_T^b
print(round(carr_madan(lambda k: a*k**b, lambda k: a*b*(b-1)*k**(b-2)), 4),
      round(D*a*S0**b*np.exp(b*(r - 0.5*sigma**2)*T + 0.5*b**2*sigma**2*T), 4))  # 10941.7428 x2
# 3) Log contract log(S_T/S0): BS value D (r - sigma^2/2) T
print(round(carr_madan(lambda k: np.log(k/S0), lambda k: -1/k**2), 6),
```

```
      round(D*(r - 0.5*sigma**2)*T, 6))       # 0.028537 0.028537
# 4) Variance swap fair strike (2 e^{rT}/T)[int P/K^2 + int C/K^2] = sigma^2 under BS
I = quad(lambda k: put(k)/k**2, 1e-8, F)[0] + quad(lambda k: call(k)/k**2, F, np.inf)[0]
print(round(2*np.exp(r*T)*I/T, 6))            # 0.04
# 5) Digital call 1{S_T >= Kb} = -dC/dK = D N(d2), via a call spread
Kb, dK = 110.0, 0.01
d2 = (np.log(S0/Kb) + (r - 0.5*sigma**2)*T) / (sigma*np.sqrt(T))
print(round((call(Kb - dK) - call(Kb))/dK, 4), round(D*norm.cdf(d2), 4))   # 0.3539 0.3539
```

2 Comparison tables

2.1 What ρ and ν do [S2–S7]

	$\rho < 0$	$\rho > 0$	$\nu \uparrow$ ($\rho = 0$)
Distribution	fat left, thin right tail (neg. skew)	fat right, thin left tail (pos. skew)	both tails fat (kurtosis)
OTM puts	up	down	up
OTM calls	down	up	up
Smile	downward skew	upward skew	curvature (both wings up)

2.2 Replication weights by payoff

Payoff $h(S_T)$	$h(F)$	$h''(K)$	Replicating portfolio
aS_T^b [S20]	aF^b	$ab(b-1)K^{b-2}$	bond aF^b, abF^{b-1} forwards, options weighted $ab(b-1)K^{b-2}$
$\log \frac{S_T}{S_0}$ [S21]	rT	$-1/K^2$	bond rT, $1/F$ forwards, short OTM options weighted $1/K^2$
S_T^2 *[Beyond slides]*	F^2	2	bond F^2, $2F$ forwards, 2 of every OTM option
$\mathbb{1}_{\{S_T \geq \bar{K}\}}$ [S28]	—	n/a	limit of call spreads: $-\partial C(\bar{K})/\partial K$

2.3 Static vs dynamic replication

	Static (Carr–Madan) [S15–S19]	Dynamic (Black–Scholes delta hedge)
Instruments	bond, forward, strip of OTM options	stock and bond
Rebalancing	none: set up once	continuous
Model needed	none (needs option prices at all strikes)	yes (dynamics and σ)
Payoffs	European, depend on S_T only	any replicable claim

3 Master formula sheet

- **Leibniz [S9]:** $\frac{\mathrm{d}}{\mathrm{d}x}\int_{u(x)}^{v(x)} f(x,t)\,\mathrm{d}t = f(x,v)v' - f(x,u)u' + \int_u^v \partial_x f\,\mathrm{d}t$.
- **Breeden–Litzenberger [S14–S15]:** $\frac{\partial C}{\partial K} = -e^{-rT}\int_K^\infty f\,\mathrm{d}s$; $\frac{\partial^2 C}{\partial K^2} = \frac{\partial^2 P}{\partial K^2} = e^{-rT}f(K)$.
- **Carr–Madan [S19]:** $V_0 = e^{-rT}h(F) + \int_0^F h''(K)P(K)\,\mathrm{d}K + \int_F^\infty h''(K)C(K)\,\mathrm{d}K$, $F = S_0e^{rT}$; uses $C' - P' = -e^{-rT}$, $C(F) = P(F)$.
- Power: $h'' = ab(b-1)K^{b-2}$. Log contract: $h(F) = rT$, $h'' = -1/K^2$.
- **Variance swap [S22–S25]:** $\sigma_R^2 = \frac{252}{N}\sum \log^2 \frac{S_{t_i}}{S_{t_{i-1}}}$; $\int_0^T \sigma_t^2\,\mathrm{d}t = 2\int_0^T \frac{\mathrm{d}S}{S} - 2\log\frac{S_T}{S_0}$; $\mathbb{E}^*\int_0^T \sigma_t^2\,\mathrm{d}t = 2e^{rT}[\int_0^F \frac{P}{K^2}\,\mathrm{d}K + \int_F^\infty \frac{C}{K^2}\,\mathrm{d}K]$.
- **VIX [S26]:** $\sigma^2 = \frac{2}{T}\sum_i \frac{\Delta K_i}{K_i^2}e^{RT}Q(K_i) - \frac{1}{T}(\frac{F}{K_0} - 1)^2$, VIX $= 100\sigma$.
- **Digital [S28–S29]:** $V_0 = -\partial C(\bar{K})/\partial K \approx [C(\bar{K} - \Delta K) - C(\bar{K})]/\Delta K$.

4 Glossary

- **Breeden–Litzenberger formula:** $\partial^2 C/\partial K^2 = e^{-rT}f(K)$ [S15]. **Butterfly spread:** discrete second difference in strike; approximates the density *[Beyond slides]*.

- **Call spread:** long low-strike, short high-strike call; approximates a digital [S29].
- **Carr–Madan formula:** static replication of a European payoff with bond, forwards and OTM options [S15, S19].
- **Cash-or-nothing digital:** pays 1 if $S_T \geq \bar{K}$ [S28]. **Implied (risk-neutral) distribution:** density implicit in option prices [S13, S15].
- **Leibniz's rule:** differentiating an integral with variable limits [S9]. **Log contract:** pays $\log(S_T/S_0)$ [S21].
- **Model-free:** prices from market option quotes, no dynamics assumed [S10]. **Static replication:** hedge set up once and held to expiry [S16].
- **Variance swap:** pays notional $\times(\sigma_R^2 - \sigma_K^2)$ [S22]. **VIX:** 100× square root of 30-day model-free implied variance [S26].

5 Self-test

Questions

Q1. ([S2–S7]) How do $\rho < 0$ and a larger ν change the return distribution and OTM option prices?

Q2. ([S9]) State Leibniz's rule.

Q3. ([S14]) Derive $\partial C/\partial K$ and $\partial^2 C/\partial K^2$ from $C(K) = e^{-rT}\int_K^\infty (s-K)f(s)\,\mathrm{d}s$.

Q4. ([S16–S19]) Derive the Carr–Madan formula. Where are put–call parity and $C(F) = P(F)$ used?

Q5. ([S20]) Replicate aS_T^b. When are the options held short?

Q6. ([S21]) Value the log contract under Black–Scholes and by static replication.

Q7. ([S23]) Show $\int_0^T \sigma_t^2\,\mathrm{d}t = 2\int \mathrm{d}S/S - 2\log(S_T/S_0)$.

Q8. ([S24–S25]) Derive the fair variance and explain the $1/K^2$ weights.

Q9. ([S28–S29]) Value a digital call with $\bar{K} \geq F$ and build a finite replication.

Solutions

S1. $\rho < 0$: fat left tail, OTM puts up, calls down (skew). $\nu \uparrow$: fatter tails both sides, OTM puts and calls up (curvature).

S2. See 1.3.

S3. $-e^{-rT}\int_K^\infty f\,\mathrm{d}s$; $e^{-rT}f(K)$ (see 1.6).

S4. See 1.8–1.9: parity gives $-C'(F) + P'(F) = e^{-rT}$ and $C(F) = P(F)$ kills the $h'(F)$ term.

S5. See 1.10; short when $0 < b < 1$ (concave payoff).

S6. $e^{-rT}(r - \frac{1}{2}\sigma^2)T$; $e^{-rT}rT - \int_0^F \frac{P}{K^2} - \int_F^\infty \frac{C}{K^2}$.

S7. Itô on $\log S$: $\mathrm{d}\log S = \mathrm{d}S/S - \frac{1}{2}\sigma^2\,\mathrm{d}t$; rearrange and integrate.

S8. See 1.13; $1/K^2$ puts more weight on low strikes, so the price depends on the skew.

S9. $-\partial C(\bar{K})/\partial K = e^{-rT}\mathbb{Q}^*(S_T > \bar{K})$; long and short $1/\Delta K$ calls at $\bar{K} - \Delta K$ and $\bar{K}$.

Appendix A. Slide coverage checklist

Slide	Content	Section
S2–S7	ρ and ν: distribution, pricing, smile	1.1–1.2
S8–S9	Differentiation of integrals; Leibniz	1.3
S10–S13	Model-free framework; implied density motivation	1.4–1.5
S14–S15	Breeden–Litzenberger; Carr–Madan statement	1.6
S16–S19	Static replication; by parts; Carr–Madan formula	1.7–1.9
S20–S21	Examples: aS_T^b; log contract	1.10–1.11
S22–S25	Variance swap contract and pricing	1.12–1.13
S26	VIX	1.14
S27–S29	Discontinuous payoffs; digital; call spread	1.15–1.16

Appendix B. Flag summary (△!)

1. [S5] "Normal log-return distribution (if $\beta = 0$)" should read $\beta = 1$.
2. [S2] ρ is said to be "proportional" to skewness; the relationship is monotone, not proportional.
3. [S25] The final variance formula uses $\mathbb{E}$; it is the risk-neutral $\mathbb{E}^*$. The fair annualized strike divides by T *[Beyond slides].*
4. [S8–S9] F denotes an antiderivative there, but the forward $S_0 e^{rT}$ from [S16] on.
5. [S21] The log-contract question has no worked answer on the slides; it is answered here with my own working.